
Improving the Sensitivity of Backdoor Detectors via Class Subspace Orthogonalization

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Abstract

Most post-training backdoor detection methods rely on attacked models exhibiting extreme outlier detection statistics for the target class of an attack, compared to non-target classes. However, these approaches may fail: (1) when some (non-target) classes are easily discriminable from all others, in which case they may *naturally* achieve extreme detection statistics (e.g., decision confidence); and (2) when the backdoor is subtle, i.e., with its features weak relative to intrinsic class-discriminative features. A key observation is that the backdoor target class has contributions to its detection statistic from both the backdoor trigger *and* from its intrinsic features, whereas non-target classes *only* have contributions from their intrinsic features. To achieve more sensitive detectors, we thus propose to *suppress* intrinsic features while optimizing the detection statistic for a given class. For non-target classes, such suppression will drastically reduce the achievable statistic, whereas for the target class the (significant) contribution from the backdoor trigger remains. In practice, we formulate a constrained optimization problem, leveraging a small set of clean examples from a given class, and optimizing the detection statistic while orthogonalizing with respect to the class’s intrinsic features. We dub this approach “class subspace orthogonalization” (CSO). CSO can be “plug-and-play” applied to a wide variety of existing detectors. We demonstrate its effectiveness in improving several well-known detectors, comparing with a variety of baseline detectors, against a variety of attacks, on the CIFAR-10, GTSRB, and TinyImageNet domains. Moreover, to make the detection problem even more challenging, we also evaluate against a novel mixed clean/dirty-label

poisoning attack that is more surgical and harder to detect than traditional dirty-label attacks. Finally, we evaluate CSO against an adaptive attack designed to defeat it, with promising detection results.

1. Introduction

Deep learning relies on large labeled training sets, making it vulnerable to backdoor attacks. Here, the adversary manipulates the training process of a deep neural network (DNN) classifier, typically via training set poisoning, so that the model learns to associate the attacker’s chosen trigger pattern with its designated target class. A backdoored model will perform normally on clean inputs, but will misclassify inputs containing the trigger to the target class. Because the poisoned fraction of the training set is small and because the trigger can be innocuous (visually subtle), such attacks can evade conventional validation, posing security risks for real-world deployment.

To address this threat, a variety of backdoor defenses have been proposed. Mitigation methods can be applied pre-training, to sanitize the training set of poisoned samples, or post-training, with model fine-tuning on clean samples used to purge a backdoor mapping from the model’s weights. The focus of this work, on the other hand, is *post-training backdoor detection*. Here, the defender *only* has access to the trained model and would like to infer whether it has been backdoored and, if so, the target class of the attack. The post-training scenario is of great significance, considering an entity (e.g., a government agency) may purchase a pre-trained AI, with no rights to access the training set. Consider also a viable small business model wherein a company fine-tunes a (possibly backdoored) pre-trained foundation model for an end-user application.

Post-training detection – without any access to the possibly poisoned training set – is quite challenging since in this case the only “evidence” for a backdoor is its imprinting on the trained weights of the network. Some of the earliest post-training methods (e.g., (Wang et al., 2019; Xiang et al., 2022)) are *reverse-engineering* detectors that assume knowledge of the mechanism (e.g., patch (Gu et al., 2019)),

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blend, additive) by which the backdoor trigger pattern is encoded into a sample. These methods are effective when the mechanism assumed by the defense is the one actually used by the attacker, but they may fail under mechanism mismatch. More recent post-training detectors do not assume knowledge of the attacker’s mechanism. (Wang et al., 2022a; 2023; Xu et al., 2024a) aim to *learn* a (UNet) model capturing the attacker’s mechanism, while MMBD (Wang et al., 2024b) hypothesizes that, *irrespective of* the mechanism used, the model tends to overfit to the backdoor trigger. While these previous works, e.g., (Wang et al., 2019; Xiang et al., 2022; Wang et al., 2022a; 2023; Xu et al., 2024a), differ in their assumptions, the *commonality* of these methods is that they are based on solving an optimization problem that yields a detection statistic for each class, with detections made when one class’s statistic is a significant outlier. In fact, these methods all try to maximize the average class posterior for a putative target class, measured over a clean set of samples originating from other classes. MMBD likewise maximizes class margin – the difference between the logit for a putative target class and the largest logit among the remaining classes – and does not exploit a clean set of samples.

While these approaches have demonstrated strong results, they can fail in two important settings. First, when the poisoning rate is quite low, but still sufficient to implant an effective backdoor, the trigger’s “signal” may be too weak to yield a significant outlier detection statistic, resulting in a missed detection. Second, when certain classes have particularly strong or unique intrinsic features, these features may produce unusually large detection statistics even in backdoor-free models, resulting in false positives. Moreover, in backdoored models, strong intrinsic features for non-target classes may yield detection statistics that rival that for the target class. In such case, the target class’s statistic will not be a significant outlier, again resulting in a missed detection.

A key observation is that the backdoor target class has contributions to its detection statistic from both the backdoor trigger *and* from its intrinsic features, whereas non-target classes *only* have contributions from their intrinsic features. Thus, to achieve more sensitive detectors, we propose to *suppress* intrinsic features while optimizing the detection statistic for a given class. For non-target classes, such suppression will drastically reduce the achievable statistic (e.g., class margin), whereas for the backdoor target class the (significant) contribution from the backdoor trigger will remain. To achieve such suppression, we propose Class Subspace Orthogonalization (CSO), a general optimization approach wherein the detector’s loss objective for a given class is altered to include a term penalizing feature directions that are not *orthogonal* to the class’s intrinsic features. CSO thus guides the detector to search in feature directions or-

thogonal to intrinsic features of the putative target class, i.e., those where the backdoor trigger’s features are more likely to “reside”. Integrating CSO into the optimization process of existing detectors helps the detector to be both more sensitive to backdoor patterns and more robust against intrinsic feature interference. Importantly, CSO is *detector-agnostic*, i.e., it can be “plug and play” applied (as seen in the sequel) to a wide range of post-training detection methods, without requiring changes to their core algorithms (excepting the modification of the detector’s loss objective). Moreover, CSO only requires a small set of clean samples for each class, which aligns well with many existing detectors that already make use of clean data.

Also, to more stringently test the effectiveness of CSO and baseline methods for backdoor detection, we also introduce a novel, mixed dirty/clean label X-to-X poisoning attack that minimizes *collateral damage* (Xiang et al., 2022), i.e., the misclassification of *non-source class*, trigger-laden samples to the target class of the attack. We note that such misclassifications are not *intended* by the attacker. In (Xiang et al., 2022), it was shown that collateral damage rates are quite high for traditional dirty label 1-to-1 (a single (source, target) class pair) attacks. To reduce these unintended misclassifications, we propose a *mixed* dirty/clean label attack, with trigger-laden samples from designated source classes (dirty-) labeled to the target class of the attack, and with trigger-laden samples from *non-source* classes (clean-) labeled to their class of origin. In addition to yielding a more “surgical” backdoor mapping, this novel attack is also substantially harder to detect than traditional dirty label attacks, as will be seen, which makes it a good, subtle attack against which to assess both CSO and existing baseline detectors.

The contributions of this paper are four-fold: (1) Our primary contribution is CSO, a novel framework that guides the detector’s optimization process toward backdoor signal directions while suppressing intrinsic class-specific signals that could otherwise dominate detection statistics. CSO is flexible and can be seamlessly integrated with existing backdoor detectors. (2) As a secondary contribution, we propose a mixed clean/dirty-label attack that applies clean-label poisoning to non-source classes. This results in an attack that is harder to detect than traditional dirty label attacks, which makes it a good challenge for both baseline and new (CSO) detectors. (3) We conduct extensive experiments on multiple benchmark datasets, demonstrating CSO’s effectiveness when combined with SOTA detectors and its robustness against adaptive attacks. (4) We comprehensively evaluate the mixed-label attack across multiple backdoor types, datasets, and network architectures, showing its ability to significantly degrade a wide range of defenses.

The rest of the paper is as follows. Section 2 develops our CSO detector. Section 3 gives experimental results. Section

4 discusses related work. Finally, Section 5 gives conclusions and directions for future work. Additional details, experiments, ablation studies and discussions are given in the appendix.

2. Methodology

2.1. Threat Model

Attacker’s Goal and Capabilities. In our mixed-label attack, the adversary has the same capabilities as in prior work (Gu et al., 2019; Wang et al., 2019; Xiang et al., 2022): the ability to (passively) poison the training set, but no access to the training process or model implementation.

Defender’s Goal and Capabilities. We consider a post-training defense setting, where the classifier has already been trained and the original training set is unavailable. The defender’s goal is to determine whether the classifier has been backdoor-attacked and to infer the attack’s target class. Following prior defenses (Wang et al., 2019; Xiang et al., 2022; Wang et al., 2023; Yang et al., 2025; Xu et al., 2024a), we assume that the defender can independently collect a small, clean dataset containing samples from all classes in the domain.

2.2. Class Subspace Orthogonalization

Generally, CSO contains *two* steps: (1) identifying the intrinsic features for each putative target class and (2) suppressing these intrinsic features during detector optimization by enforcing orthogonality to the class’s intrinsic subspace.

2.2.1. MOTIVATION FOR THE CASE OF A MAXIMUM LOGIT DETECTOR

Consider a K -class problem, working in the space of augmented feature vectors $\mathbf{y}^T = (\mathbf{x}^T, 1)$. Suppose the class set is $\mathcal{K}$, and for each class, $k \in \mathcal{K}$, there is a linear discriminant function $g_k(\mathbf{y}) = \mathbf{w}_k^T \mathbf{y}$. The classifier employs a winner-take-all decision rule: $\hat{c}(\mathbf{y}) = \arg \max_j g_j(\mathbf{y})$. For class k , the clean training set is $\mathcal{Y}_k = \{\mathbf{y}_1^{(k)}, \dots, \mathbf{y}_{N_k}^{(k)}\}$. Suppose the training set for source class s is dirty-label backdoored with target class t . For non-target classes $k \neq t$, we (reasonably) assume the weight vector can be expressed in the form $\mathbf{w}_k = \sum_{i=1}^{N_k} \gamma_i^{(k)} \mathbf{y}_i^{(k)}$, where $\gamma_i^{(k)} \geq 0, \forall i$, i.e., it is a non-negative linear combination of the training vectors. For the target class t , we write $\mathbf{w}_t = \sum_{i=1}^{N_t} \gamma_i^{(t)} \mathbf{y}_i^{(t)} + \alpha \mathbf{b}^{(t)}$, where $\gamma_i^{(t)} \geq 0, \forall i, \alpha > 0$, and $\mathbf{b}^{(t)}$ is a sample containing the backdoor pattern. We will suppose that $\mathbf{b}^{(t)} \notin \text{span}(\mathcal{Y}_t)$. This is reasonable, given that the backdoor trigger *modifies* samples from class $s \neq t$. For example, in an *additive* attack mechanism, $\mathbf{b}^{(t)} = \mathbf{y}^{(s)} + \mathbf{b}$, $\mathbf{y}^{(s)}$ a source-class sample and $\mathbf{b}$ the backdoor pattern. Clearly, in general, $\mathbf{y}^{(s)} \notin \text{span}(\mathcal{Y}_t)$

and, thus, $\mathbf{b}^{(t)} \notin \text{span}(\mathcal{Y}_t)$.

Baseline Maximum-Logit Backdoor Detection (MLBD).

A maximum-logit detector determines, for each class k :

$$l_k = \max_{\mathbf{y}} \mathbf{w}_k^T \mathbf{y} \quad \text{s.t.} \quad \|\mathbf{y}\|_2 = 1, \quad (1)$$

with the constraint needed to make the problem well-posed. The detector then flags a backdoor if $l_{k^*} = \max_k l_k$ is an outlier. However, if a non-target class $\tilde{t} \neq t$ has strong intrinsic features such that $l_{\tilde{t}} \approx l_t$ or even $l_{\tilde{t}} > l_t$, l_t may not be an outlier, leading to a missed detection.

Orthogonalized MLBD. To resolve this issue, we modify the optimization by adding class-subspace orthogonality constraints:

$$l_k = \max_{\mathbf{y}} \mathbf{w}_k^T \mathbf{y} \quad \text{s.t.} \quad \|\mathbf{y}\|_2 = 1, \mathbf{y}^T \mathbf{y}_i^{(k)} = 0, i = 1, \dots, N_k. \quad (2)$$

That is, we must have: $\mathbf{y} \perp \text{span}(\mathcal{Y}_k)$. For $k \neq t$, since $\mathbf{w}_k \in \text{span}(\mathcal{Y}_k)$, we have $\mathbf{w}_k^T \mathbf{y} = 0$ for all feasible $\mathbf{y}$, yielding $l_k = 0$. For $k = t$, we can write $\mathbf{w}_t = \mathbf{w}_t^{\text{benign}} + \mathbf{b}_\perp^{(t)}$, where $\mathbf{w}_t^{\text{benign}} \in \text{span}(\mathcal{Y}_t)$, and $\mathbf{b}_\perp^{(t)}$ is the component of $\alpha \mathbf{b}^{(t)}$ orthogonal to $\text{span}(\mathcal{Y}_t)$. The orthogonality constraint removes $\mathbf{w}_t^{\text{benign}}$, leaving:

$$l_t = \max_{\mathbf{y}} \mathbf{b}_\perp^{(t)T} \mathbf{y} \quad \text{s.t.} \quad \|\mathbf{y}\|_2 = 1, \quad (3)$$

whose solution is $\mathbf{y}^* = \mathbf{b}_\perp^{(t)} / \|\mathbf{b}_\perp^{(t)}\|$ with $l_t = \|\mathbf{b}_\perp^{(t)}\| > 0$. Thus, via orthogonalization, all l_k vanish for $k \neq t$ while l_t remains positive, making l_t a clear outlier. This suggests orthogonalization can be the basis for improved detection sensitivity.

2.2.2. IDENTIFYING INTRINSIC, CLASS-SPECIFIC FEATURES

For purposes of backdoor detection, let the DNN classifier be $f(\cdot) = S_b \circ S_a(\cdot)$, where S_a denotes the feature extractor (e.g., convolutional layers) and S_b performs back-end classification. For an input $\mathbf{x}$ in the input space $\mathcal{X}$, the internal feature vector is $S_a(\mathbf{x}) \in \mathbb{R}^d$. Inspired by feature decoupling methods (Wang et al., 2019; 2022a; 2023; Xu et al., 2024a), for each class k , we aim to learn a soft mask $\mathbf{v}_k \in [0, 1]^d$ that identifies its intrinsic feature subspace. Given a small clean set for class k , $\mathcal{D}_k = \{\mathbf{x}_i^{(k)}, i = 1, \dots, M_k\}$, we choose $\mathbf{v}_k$ to (simultaneously) minimize classifier loss on the intrinsic features and maximize classifier loss on the complement set:

$$\min_{\mathbf{v}_k} \sum_{\mathbf{x}^{(k)} \in \mathcal{D}_k} \left[\mathcal{L}(S_b(S_a(\mathbf{x}^{(k)}) \odot \mathbf{v}_k), k) - \mathcal{L}(S_b(S_a(\mathbf{x}^{(k)}) \odot (1 - \mathbf{v}_k)), k) \right], \quad (4)$$

where $\odot$ is the element-wise (Hadamard) product, and $\mathcal{L}$ is, e.g., the cross-entropy loss. This produces an intrinsic subspace for each class k : $\text{span}(\mathbf{v}_k \odot S_a(\mathbf{x}^{(k)}) : \mathbf{x}^{(k)} \in \mathcal{D}_k)$.

Note that, unlike existing feature decoupling methods that use soft-masking in activation space (Wang et al., 2022a; 2023; Xu et al., 2024a), we are estimating a *class-specific* mask. This is both reasonable and necessary since the features that are class-discriminating will in general be class-specific.

2.2.3. ORTHOGONALIZATION IN DETECTOR OPTIMIZATION

Let Θ be the variables over which the detector optimizes (e.g., the DNN’s input, or parameters specifying how a putative backdoor trigger is incorporated into a sample, in the case of a reverse engineering based detector). We write $z(\Theta; x)$ to represent a putative backdoor-triggered sample induced by Θ (potentially acting on an input sample, x), with $S_a(z(\Theta; x))$ its internal features.

For each putative target class t , exploiting the learned class-specific mask v_t , to effectively orthogonalize, we penalize positive correlations between $S_a(z(\Theta))$ and the intrinsic subspace for class t . That is, we define the CSO penalty for $z(\Theta; x)$ as the average rectified cosine similarity $C_t(z(\Theta; x)) =$

$$\frac{1}{|D_t|} \sum_{x^{(t)} \in D_t} \text{ReLU} \left(\frac{\langle S_a(z(\Theta; x)), v_t \odot S_a(x^{(t)}) \rangle}{\|S_a(z(\Theta; x))\| \|v_t \odot S_a(x^{(t)})\|} \right), \quad (5)$$

with inner (dot) product $\langle \cdot, \cdot \rangle$. This novel penalty will be used to discourage alignment with intrinsic features of a putative target class, pushing the detector’s search into subspaces more likely to contain backdoor-related components, and with the ReLU operation leaving *anti-correlated* directions unaffected.

2.3. Detectors + CSO

Let $\mathcal{J}_t(\cdot)$ denote the objective function of a given detector for putative target class t . We next develop CSO variants of several well-known detectors.

2.3.1. MMBD-CSO

MMBD (Wang et al., 2024b) exploits an empirically observed backdoor overfitting phenomenon, searching for an input that maximizes the *margin*, i.e., the (non-negative) difference between the logit for putative target class t and the second-largest logit. In this case, since the detector directly optimizes over the input space, we have $z(\Theta; x) = z \in \mathcal{X}$. Incorporating the CSO penalty term, we have the following MMBD-CSO loss objective:

$$\mathcal{J}_t(z) = -[g_t(z) - \max_{k \in \mathcal{K} \setminus t} g_k(z)] + \lambda C_t(z), \quad (6)$$

with λ giving the relative weight to the CSO penalty. The detection statistic in this case is the achieved maximum margin. Note that the original MMBD approach had no way

to benefit from any available clean samples. MMBD-CSO exploits clean samples, and as will be seen, significantly improves on the detection performance of MMBD.

We also consider Maximum Logit Backdoor Detection (MLBD)-CSO, with the margin term replaced by the logit:

$$\mathcal{J}_t(z) = -g_t(z) + \lambda C_t(z). \quad (7)$$

Here the detection statistic is the achieved maximum logit.

2.3.2. NC-CSO

Neural Cleanse (Wang et al., 2019) assumes a *blended* attack mechanism, defined by a *spatial* soft mask $m_{\text{NC}} \in [0, 1]^{\dim(\mathcal{X})}$ and trigger pattern $p \in \mathcal{X}$, i.e., $\Theta = \{m_{\text{NC}}, p\}$ and $z(\Theta; x) = (1 - m_{\text{NC}}) \odot x + m_{\text{NC}} \odot p$. The Neural Cleanse detector jointly chooses the spatial mask and pattern to effectively maximize the fraction of non-target class clean samples induced, by the blending operation, to be classified to the putative target class, t . With the incorporation of a CSO penalty, the NC-CSO loss objective $\mathcal{J}_t(\{m_{\text{NC}}, p\}) =$

$$\sum_{x^{(k)} \in \mathcal{D}_k, k \neq t} \mathcal{L}(f((1 - m_{\text{NC}}) \odot x^{(k)} + m_{\text{NC}} \odot p), t) + r^* + \lambda \sum_{x^{(k)} \in \mathcal{D}_k, k \neq t} C_t((1 - m_{\text{NC}}) \odot x^{(k)} + m_{\text{NC}} \odot p). \quad (8)$$

where $\mathcal{L}$ is the cross entropy loss and r^* is a regularization term (e.g., to encourage small trigger size and small mask size). So, NC-CSO soft-masks *both* the input and the embedded (S_a) feature space. The soft-masking of the input (image) estimates the *spatial support* of the backdoor trigger, while soft-masking in activation space estimates the class’s intrinsic features, defined on this spatially localized support. These two feature maskings are, thus, complementary, and both are necessary within NC-CSO. Here, the detection statistic is the estimated mask norm $\|m_{\text{NC}}\|$, where a smaller mask norm serves as evidence of a potential backdoor target class.

2.3.3. PT-RED-CSO

PT-RED (Xiang et al., 2022) seeks the smallest additive perturbation $p \in \mathcal{X}$ that causes most inputs from a putative source class s to be misclassified to a putative target class t , i.e., in this case $\Theta = \{p\}$ and $z(\Theta; x) = x + p$. With a CSO penalty incorporated, the PT-RED-CSO loss objective becomes:

$$\mathcal{J}_{s,t}(p) = \sum_{x^{(s)} \in \mathcal{D}_s} \mathcal{L}(f(x^{(s)} + p), t) + \lambda \sum_{x^{(s)} \in \mathcal{D}_s} C_t(x^{(s)} + p), \quad (9)$$

with $\mathcal{L}$ the cross entropy loss. The detection statistic is the perturbation norm $\|p\|$, with smaller norms indicating higher likelihood of a backdoor mapping from $s \rightarrow t$.

Note that these 4 CSO variants differ both in the detector objective function and in the variables being optimized over.

Thus, these 4 variants give a good idea about the “plug-and-play” generality of the CSO framework. More generally, so long as an existing detector is optimizing a detection statistic that relies on intrinsic feature contributions, CSO should in principle be applicable to this existing detector. The detection benefits of such inclusion will be seen shortly.

2.4. Mixed Label Attack

Next, we introduce a novel, stealthy attack that will be used in our experiments as a significant detection challenge, both for CSO detection methods and for the baseline methods with which we will compare. In an X-to-X attack, let the attacker’s poisoned source class set be $\mathcal{S} \subseteq \mathcal{K}$ and the target class set be $\mathcal{T} \subseteq \mathcal{K}$. In a traditional dirty-label attack, the attacker mislabels a subset of samples from $\mathcal{S}$ to $\mathcal{T}$ and allows the model to be trained as usual. However, in our approach, the attacker performs an additional step by *also* embedding the backdoor trigger into samples from $\mathcal{K} \setminus (\mathcal{S} \cup \mathcal{T})$ while keeping their original labels intact. This teaches the model to learn to *only* misclassify to $\mathcal{T}$ when the backdoor trigger is applied to samples from $\mathcal{S}$, i.e., so that there is little to no collateral damage.

To characterize this attack, we introduce the following metrics: 1) Dirty Poisoning Rate (DPR): the number of poisoned samples from $\mathcal{S}$ mislabeled to $\mathcal{T}$, divided by the total number of training samples; 2) Clean Poisoning Rate (CPR): the number of clean-label poisoned samples divided by the total number of training samples; 3) Overall Poisoning Rate (OPR): the sum of DPR and CPR. In traditional dirty-label attacks, $\text{OPR} = \text{DPR}$. In contrast, we introduce additional poisoned but correctly labeled samples, making $\text{DPR} < \text{OPR}$. Similar to other attacks, at modest poisoning rates, our attack achieves i) a high attack success rate on samples from the intended source classes when the backdoor trigger is applied and ii) minimal degradation in clean test accuracy. However, unlike traditional attacks, our attack also greatly reduces collateral damage. Moreover, as will be seen, this attack is much harder to detect than conventional dirty-label poisoning and, thus, represents a good challenge, both for CSO and baseline detection methods.

3. Experimental Results

3.1. Experiment Setup

Dataset and models. We experiment on three benchmark datasets: CIFAR-10 (Krizhevsky & Hinton, 2009), GTSRB (Houben et al., 2013), and (a subset of) TinyImageNet (Le & Yang, 2015), containing 40 classes. We evaluate our methods for ResNet-18 (He et al., 2016), PreActResNet-18 (Yu et al., 2018), VGG-16 (Simonyan & Zisserman, 2015), and ViT (Dosovitskiy et al., 2021) on CIFAR-10, MobileNet (Howard et al., 2017) on GTSRB, and ResNet-34 on Tiny-

ImageNet. Results on VGG-16 and ViT are reported in Appendix H. We randomly select 10 clean training samples per class for use by our CSO variants. Please see Appendix A.1 for more dataset details.

Backdoor Attacks. We assess against sixteen representative and advanced backdoor attacks: (1) BadNets (Gu et al., 2019), (2) chessboard (Xiang et al., 2022), (3) 1-pixel (Tran et al., 2018), (4) blend (Wang et al., 2019), (5) WaNet (Nguyen & Tran, 2021), (6) input-aware (dubbed ‘IAD’) (Nguyen & Tran, 2020), (7) label-consistent (dubbed ‘LC’) (Turner et al., 2019), (8) Bpp (Wang et al., 2022b), (9) Re-fool (Liu et al., 2020), (10) Narcissus (Zeng et al., 2023), (11) SIG (Barni et al., 2019), (12) Bypass (Shokri et al., 2020), (13) DRUPE (Tao et al., 2024), (14) MTBA (Li et al., 2024), (15) multi-target (Xue et al., 2023) and (16) TITIM (Lin et al., 2025). Attacks (9)-(16) are evaluated in Appendix F. LC is not reported on TinyImageNet as the attack fails despite high poisoning rates. For the mixed dirty/clean-label attack (which we denote by ‘ML’ (mixed labeling) in the sequel), we consider both one-to-one and multi-to-one cases on BadNets, chessboard, 1-pixel and blend. CPR was set equal to DPR, i.e. a very modest amount of clean label poisoning – e.g., for BadNet both DPR and CPR were 0.1%. We experimented under a *lowest* feasible poisoning rate scenario. That is, the adversary chooses the *minimum* poisoning rate needed to achieve a high attack success rate. *This makes the detection problem most challenging.* Detailed attack settings—including the poisoning rate (PR), attack success rate (ASR), clean test accuracy (ACC), and, for one-to-one and multi-to-one attacks, collateral damage (CD)—are reported in Appendix A.3.

Backdoor Defenses. We tested five well-known backdoor defense methods as baselines, i.e., Neural Cleanse (NC) (Wang et al., 2019), PT-RED (Xiang et al., 2022), MMBD (Wang et al., 2024b), UNICORN (Wang et al., 2023), and BTI-DBF (Xu et al., 2024a), compared against CSO variants of NC, PT-RED, MMBD, and MLBD. We closely adhered to the original implementations of all baseline methods. However, BTI-DBF always detects models as poisoned, that is, it gives 100% false positives on clean models. In order to reduce false positives, we endow BTI-DBF with a detection rule, applying a threshold to their maximum class proportion measure. We report under two thresholds: $1/|\mathcal{K}|$ (favoring backdoor detection) and 1.0 (minimizing false positives), respectively denoted BTI-DBF and BTI-DBF-2. UNICORN was excluded from the TinyImageNet evaluation due to prohibitive runtime. Additional implementation details for all baselines are provided in Appendix A.2. For the CSO variants, we chose $S_a()$ as the last convolution layer. Justification for this choice is given in Appendix B.1. We set λ to achieve approximate balance between the detector objective and the CSO penalty. This translates to setting $\lambda = 0.01$ for NC, $\lambda = 0.1$ for PT-RED, and $\lambda = 400$ for

MMBD-CSO and MLBD-CSO.

Evaluation Metrics. We adopt detection accuracy to evaluate the performance for the considered detectors and also the effectiveness of the mixed label attack, compared with the other attacks. A successful detection requires both the backdoor attack to be detected and the target class to be correctly inferred. Clean models are used to evaluate the false positive detection rate.

3.2. Backdoor Detection Performance

For all datasets, we trained 10 clean models and 10 backdoored models with randomly chosen target classes and applied all detection methods for each model 5 times. (All the detectors involve random initialization for their optimization, so we repeated the experiment for 5 different seeds.) We calculate detection accuracy % (DA) over all 50 trials. For clean models, one minus the false positive rate is reported. The number of clean images per class is reported as N_{img} . Detection results where the CSO variant improves over its corresponding baseline by at least 20% are highlighted in gray, and the best performance for each attack type is shown in bold.

The results are shown in Tables 1, 2, 3, 4, with the poisoning rates for dirty-label attacks and dirty poisoning rates for mixed-label attacks reported in parentheses. Surveying these tables, we make the following key observations: 1) CSO variants substantially outperform their baselines. For example, MMBD-CSO achieves much higher DA than MMBD in Table 2, *across all attacks*. Likewise, CSO variants of PT-RED and NC substantially outperform their baselines; 2) CSO variants have low false positives, particularly MMBD-CSO. It achieves just 4% in Tables 2, 14% in Table 3 and 12% in Table 4. By comparison, UNICORN and BTI-DBF give much higher false positives (even though BTI-DBF uses many more (e.g. 250 for CIFAR-10) clean images than the CSO methods); 3) MMBD-CSO is the overall best-performing detector, across these experiments, with both higher (in many cases much higher) DAs overall and lower false positives overall than all other detectors; 4) MLBD-CSO fares worse than MMBD-CSO, which suggests that margin, rather than logit, maximization yields a more robust detection statistic; 5) One-to-one and multi-to-one attacks are consistently harder to detect than all-to-one attacks at similar poisoning rates (PRs), as reflected by lower DA across datasets and detectors. Moreover, surveying Table 2, 3 and 4, ML attacks are much harder to detect than the baseline dirty-label attacks (even though CPR=DPR, *i.e.* the amount of clean label poisoning is very low). For example, in Table 2, with very few exceptions, ML detection accuracies are *substantially* lower than baselines, across all attack types, and against all detectors. This also holds true for the seven-to-one attacks in Tables 3 and 4. For example,

MMBD-CSO’s detection accuracy drops from 96% for 1-pixel to 68% for ML-1-pixel in Table 3. 6) For these low poisoning rate experiments, detection rates are much lower than reported in original papers, where higher poisoning rates were used. For example, UNICORN reported 95% detection accuracy for BadNet with 5% poisoning rate, but with 0.1% poisoning rate the detection accuracy is only 54%; BTI-DBF reported 100% detection accuracy with 5% poisoning rate, but with 0.1% poisoning rate the detection accuracy is only 72%. MMBD and MMBD-CSO both achieve much higher detection rates at a higher poisoning rate, as shown in Appendix I.

3.3. Adaptive Attack

We further examine whether our CSO defense can be compromised when the adversary has knowledge of the method. Class subspace feature decoupling and CSO leverage the assumption that backdoor and intrinsic features are separable in the latent space. To exploit this, we introduce a new adaptive attack, Adaptive-Blend, designed to undermine CSO by choosing the backdoor trigger to contain intrinsic target class features. For CIFAR-10, we consider ‘dog’ as the backdoor target class, with the backdoor pattern a 16×16 pixel dog’s face. For each poisoned training image, we randomly crop the backdoor to a 8×8 patch and blend it into the source class image. By using a different random cropping for each poisoned image, we are capturing *all* of the dog’s intrinsic features, across the collection of poisoned training images. At the same time, we are limiting the backdoor pattern to 8×8 (16×16 would be too large, given the small size of CIFAR-10 images). At test time, we evaluate the attack success rate by blending an 8×8 (dog) backdoor trigger pattern into test samples. We consider blending ratios from 0.2 all the way up to 1.0 (a patch attack).

In Table 5, we show the trigger-intrinsic overlap, which is the average rectified cosine similarity between the backdoored source class image features (masked to capture intrinsic features of the target class) and the target class clean image features. We also show the target class intrinsic feature overlap, which is the average rectified, masked feature cosine similarity between target class clean samples as a baseline for comparison. The trigger-intrinsic overlap becomes larger as the blend ratio increases. This is as one would expect. However, MMBD-CSO *still* achieves excellent detection accuracy, even up to a blend ratio of 1.0. Even with CSO’s intrinsic feature penalization, MMBD-CSO may still rely on intrinsic target-class features when these are the only features capable of producing a nonzero margin. Increasing the CSO penalty λ from 400 to 400000 and evaluating attacks with blend ratios of 0.9, 0.95 and 1.0 results in detection accuracies of 78%, 54% and 36%, respectively. This suggests that extraordinarily large CSO penalties can degrade performance at high blend ratios. No-

Improving the Sensitivity of Backdoor Detectors via Class Subspace Orthogonalization

Table 1. Detection accuracies for CIFAR-10 under 8 different all-to-one backdoors. Highlighted cells indicate $\geq 20\%$ relative improvement of CSO over its baseline. Bold indicates best detector for a given attack. Note that very low poisoning rates (shown in parentheses), chosen to make the detection problem most challenging, were used in this experiment.

CIFAR-10		All-to-one attack (Poisoning Rate (%))								
Detector	N_{img}	clean (0)	BadNet (0.1)	chess (0.5)	1-pixel (0.5)	blend (5.0)	WaNet (5.0)	IAD (5.0)	LC (5.0)	Bpp (1.0)
UNICORN	10	52	54	52	38	76	56	34	74	40
BTI-DBF	250	0	72	100	100	100	34	32	4	6
BTI-DBF-2	250	40	12	60	76	48	20	32	0	0
NC	10	72	90	100	100	82	26	44	0	0
NC-CSO	10	70	100	100	100	100	54	74	4	18
PT-RED	100	62	0	100	100	90	46	4	10	10
PT-RED-CSO	100	72	54	100	100	100	62	12	10	10
MMBD	0	86	82	100	100	98	90	76	30	0
MMBD-CSO	10	96	92	100	100	100	88	84	76	36
MLBD-CSO	10	96	90	100	100	100	84	66	56	32

Table 2. Detection accuracies for CIFAR-10 under 11 one-to-one backdoors. Dirty-label and mixed-label (ML) settings are reported.

CIFAR-10		One-to-one attack (Poisoning Rate/Dirty Poisoning Rate (%))											
Detector	N_{img}	clean (0)	BadNet (0.1)	ML-BadNet (0.1)	chess (0.2)	ML-chess (0.2)	1-pixel (3.0)	ML-1-pixel (3.0)	blend (2.0)	ML-blend (2.0)	WaNet (5.0)	IAD (3.0)	Bpp (1.0)
UNICORN	10	52	80	56	28	22	80	42	28	22	18	28	54
BTI-DBF	250	0	84	60	86	52	100	84	74	74	24	60	46
BTI-DBF-2	250	40	42	30	60	44	66	36	44	10	0	0	0
NC	10	72	94	48	70	42	78	10	60	48	0	28	0
NC-CSO	10	70	90	66	82	54	96	38	76	74	14	56	8
PT-RED	100	62	10	12	100	90	100	84	90	78	10	40	0
PT-RED-CSO	100	72	12	16	100	100	100	92	98	90	10	60	12
MMBD	0	86	66	40	32	0	68	20	52	12	0	10	0
MMBD-CSO	10	96	96	70	78	66	100	94	84	78	20	48	36
MLBD-CSO	10	96	88	46	78	46	100	46	78	72	18	48	30

Table 3. Detection accuracies for GTSRB under 8 all-to-one backdoors and 8 seven-to-one backdoors.

GTSRB		All-to-one attack (Poisoning Rate (%))										Seven-to-one attack (Poisoning Rate/Dirty Poisoning Rate (%))							
Detector	N_{img}	clean (0)	BadNet (0.1)	chess (1.1)	1-pixel (1.1)	blend (0.5)	WaNet (5.0)	IAD (5.0)	LC (3.8)	Bpp (1.0)	BadNet (3.2)	ML-BadNet (3.2)	chess (8.1)	ML-chess (8.1)	1-pixel (3.2)	ML-1-pixel (3.2)	blend (3.2)	ML-blend (3.2)	
UNICORN	10	46	12	10	34	32	82	42	22	46	98	42	50	38	84	34	98	90	
BTI-DBF	60	0	18	18	36	40	36	12	20	76	88	74	78	20	80	48	100	62	
BTI-DBF-2	60	0	18	18	34	38	36	12	0	0	76	70	60	20	78	52	50	40	
NC	10	62	72	90	72	54	72	10	42	100	90	44	44	18	40	44	60	58	
NC-CSO	10	74	82	100	82	90	78	10	44	100	92	60	64	30	72	50	82	66	
PT-RED	10	42	44	78	60	70	80	0	52	92	22	18	34	20	36	40	56	30	
PT-RED-CSO	10	66	42	90	88	88	92	18	56	86	36	20	54	40	62	54	66	56	
MMBD	0	76	52	10	60	78	88	24	100	100	70	40	18	10	72	38	52	68	
MMBD-CSO	10	86	78	48	100	98	100	36	100	100	92	58	80	44	96	68	100	92	

Table 4. Detection accuracies for TinyImageNet under 8 all-to-one backdoors and 8 seven-to-one backdoors.

TinyImageNet		All-to-one attack (Poisoning Rate (%))										Seven-to-one attack (Poisoning Rate/Dirty Poisoning Rate (%))					
Detector	N_{img}	clean (0)	BadNet (1.0)	chess (3.0)	1-pixel (5.0)	blend (10.0)	WaNet (5.0)	IAD (5.0)	Bpp (1.0)	BadNet (1.8)	ML-BadNet (1.8)	chess (5.3)	ML-chess (5.3)	1-pixel (5.3)	ML-1-pixel (5.3)	blend (14.0)	ML-blend (14.0)
BTI-DBF	25	68	24	86	90	100	12	16	4	28	40	86	28	82	50	94	66
BTI-DBF-2	25	82	10	40	70	78	12	0	4	18	10	78	0	60	0	56	0
NC	10	10	100	74	76	86	38	0	30	100	84	96	52	44	30	50	52
NC-CSO	10	28	100	84	70	100	36	12	44	100	100	100	58	62	48	76	50
PT-RED	10	100	40	88	72	62	28	14	0	50	40	100	64	44	32	64	56
PT-RED-CSO	10	100	56	100	100	100	48	10	12	64	60	100	66	70	54	70	68
MMBD	0	78	96	28	94	100	100	0	62	96	60	18	0	60	30	50	46
MMBD-CSO	10	88	98	48	100	100	100	28	100	100	66	68	30	88	68	100	80

tably, MMBD-CSO does not fundamentally fail on adaptive attacks; degradation in detection accuracy occurs only under

penalty settings far beyond practical use. In Appendix C, we provide an illustration of our Adaptive-Blend attack and

Table 5. Results for Adaptive-Blend with different blend ratios.

Blend ratio→	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	0.95	1.0
Trigger-Intrinsic Feature Overlap	0.40	0.40	0.54	0.52	0.60	0.53	0.64	0.57	0.62	0.65
Target Class Intrinsic Feature Overlap	0.67	0.63	0.65	0.65	0.68	0.64	0.67	0.62	0.64	0.66
ASR(%)	88	90	97	98	99	100	100	100	100	100
ACC(%)	91	91	91	91	91	91	91	91	91	91
DA(%)	100	100	100	100	96	96	90	92	92	90

Table 6. DA for different split positions.

Attack↓, layer→	9 th	11 th	13 th	15 th	Last
clean	88	90	96	98	96
BadNet	74	78	88	90	96
ML-BadNet	56	58	64	64	70

Table 7. DA for different CSO penalty values.

Attack↓, $\lambda \rightarrow$	100	200	400	600	800
clean	84	90	96	96	90
BadNet	70	90	96	92	96
ML-BadNet	44	66	70	70	68

further evaluate against Adaptive-Blend-2 (Qi et al., 2023).

3.4. Ablation Studies

The defense considered here is MMBD-CSO, and the attacks are one-to-one BadNet and ML-BadNet. We use 10 clean and 10 backdoored ResNet-18 models on CIFAR-10 to assess detection results.

Effects of Split Position for Feature Decoupling. Table 6 shows the DA results of using different depths of the feature extractor S_a – the 9th, 11th, 13th, 15th, and last convolution layers. As seen, splitting at later layers is generally better than splitting at earlier layers.

Effects of CSO constraint penalty hyperparameter. In Table 7, we evaluate the impact of λ . We observe that performance is not very sensitive, so long as λ is not made too small. This is plausible because a larger λ implies a greater degree of class subspace orthogonalization.

3.5. Time Complexity

Figure 1 reports the execution time on CIFAR-10/ResNet-18 for the detectors compared in this work. The CSO variants introduce only a modest overhead compared to their baselines.

In Appendix E, we provide additional details on the time complexity, memory usage and efficiency ratio of CSO.

3.6. Additional Experimental Results

Appendix B.1 shows that triggers are in general only weakly correlated with intrinsic features of the target class, which supports use of CSO. It also shows how this correlation depends on the network layer. Appendix B.2 shows that there are in fact significant feature correlations between classes –

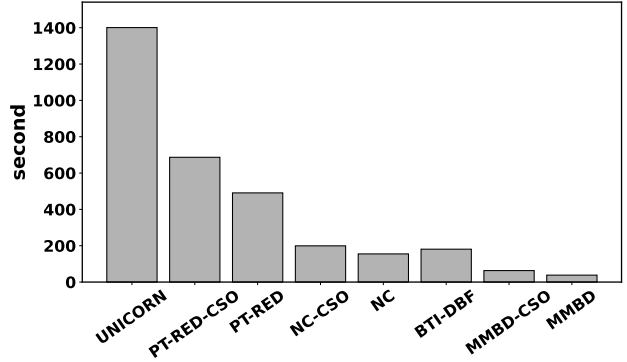


Figure 1. Time complexity of the considered detectors.

in spite of this, CSO methods are quite successful at detection. Appendix D includes additional ablation studies that show the effect of N_{img} , the need for class-specific feature masking, for applying the ReLU to the cosine similarity, and the robustness of CSO when the clean set used for detection is subject to domain shift or to mislabeling. In Appendix F, we evaluate MMBD-CSO against additional stealthy attacks that aim to synthesize backdoor signals to closely resemble clean samples in the embedded feature space, clean label attacks, multi-trigger attacks, multi-target attacks and TITIM attacks. In Appendix G, we compare CSO with another recent detector BAN (Xu et al., 2024b). We also evaluate MMBD-CSO against larger DNN architectures VGG-16 and ViT in Appendix H. In Appendix I, we evaluate MMBD-CSO against WaNet, IAD, and LC at a higher poisoning rate. These results (with high detection rates) show that the low detection rates for these attacks in Tables 1 and 2 are largely due to the use of an unusually low poisoning rate. We incorporate CSO into backdoor mitigators in Appendix J, which shows that CSO not only improves backdoor detectors but also helps backdoor mitigators. In Appendix K, we discuss in detail how our mixed-label attack affects collateral damage and detection performance. In Appendix L, we discuss the backdoor’s effects on class logits and the reason why MMBD-CSO outperforms MLBD-CSO. Finally, Appendix M discusses the limitations of our method and outlines possible directions for future work.

4. Related Work

4.1. Backdoor Defense

Feature masking in backdoor detection has been widely applied, as early as (Liu et al., 2018; Wang et al., 2019), and more recently in (Huang et al., 2022; Wang et al., 2022a; 2023; Xu et al., 2024a), and our use of it is inspired by these earlier works. However, unlike (Wang et al., 2022a; 2023; Xu et al., 2024a), our approach learns a *class-dependent* soft feature mask, in order to identify a putative target class’s intrinsic, discriminating features. Note also that our method,

using class-dependent masks, outperforms (Huang et al., 2022; Wang et al., 2022a; 2023; Xu et al., 2024a), which use global masks. The necessity of using a class-dependent mask for CSO is demonstrated in Appendix D.

We first came up with the CSO approach by considering detectors such as MMBD (Wang et al., 2024b), which have no mechanism for exploiting an available, small clean set of samples. We sought to redress that limitation. While CSO gives such a mechanism, we further recognized its wider applicability, beyond just improving MMBD. While cosine similarity has been part of a backdoor defense previously (Zeng et al., 2024), its use there was to effectively encourage clustering in activation space – quite different from class subspace orthogonalization.

4.2. Backdoor Attacks

Clean-label attacks are stealthier than dirty-label attacks in that they do not require any mislabeling. This may be accompanied by strategies encouraging the model to associate the trigger with the target class (Turner et al., 2018; Zhao et al., 2020; Souri et al., 2022). We use clean-label poisoning for a different purpose – not to make source classes trigger-susceptible, but to eliminate “collateral damage”. Moreover, as shown here, even with very modest CPR, this attack is much harder to detect than a pure dirty-label attack. Also, unlike adaptive dirty-label attacks or sophisticated clean-label strategies (Turner et al., 2018; Zhao et al., 2020; Souri et al., 2022), our attack is *passive*: it requires only a poisoning capability, without assuming the attacker is the training authority, has surrogate model access, or access to a large amount of data from the domain.

5. Conclusions

In this work, we developed a general “plug-and-play” framework – class subspace orthogonalization (CSO) – for enhancing the sensitivity of backdoor detectors. We evaluated CSO-enhanced detectors and baselines against a variety of existing attacks, as well as against a novel mixed dirty/clean label attack, proposed here, that is significantly harder to detect than traditional dirty-label attacks. Evaluation on the mixed label attack shows that existing detectors are inadequate, and CSO is necessary, to help detect subtle, stealthy threats. In future, we may consider whether CSO can also benefit existing backdoor *mitigation* techniques.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. More Details of Experiment Settings

A.1. Datasets

CIFAR10 (Krizhevsky & Hinton, 2009). This dataset is built for recognizing general objects such as cats, deer, and automobiles. It contains 50,000 training samples and 10,000 test samples in 10 classes.

GTSRB (Houben et al., 2013). The German Traffic Sign Recognition Benchmark is designed for traffic sign classification. It contains 39,209 training images and 12,630 test images across 43 classes of traffic signs.

TinyImageNet (Le & Yang, 2015). This dataset is a scaled-down version of the ImageNet (Russakovsky et al., 2015) benchmark. It consists of 100,000 training samples and 10,000 validation samples over 200 object categories, each with 500 training images and 50 validation images. In this paper, we select a subset of the original TinyImageNet which contains the first 40 classes.

A.2. Settings for detection methods

In this section, we describe the detailed settings of the detectors used in our experiments.

UNICORN (Wang et al., 2023). We follow the default settings used in its original paper. Also, we determine a model is clean, i.e., without a backdoor, if the ASR-Inv of all labels is no larger than 90%.

BTI-DBF (Xu et al., 2024a). We follow the default settings used in its original paper. BTI-DBF *always* detects a backdoor is present since it predicts the target label by selecting the class with the highest frequency over the model’s predictions. In order to endow detection specificity to BTI-DBF, we define a threshold for this frequency ratio. Specifically, for BTI-DBF results reported in our experiments, if the maximum frequency ratio exceeds $1/|\mathcal{K}|$ (the uniform random baseline over $|\mathcal{K}|$ classes), it is considered an indicator of a potential backdoor. In BTI-DBF-2, we instead set this threshold to 1.0. The former choice prioritizes backdoor detection sensitivity, while the latter aims to minimize false positives in clean models.

NC (Wang et al., 2019). We follow the default settings used in its original paper.

NC-CSO. We set $\lambda = 0.01$ in Eq. 8. All remaining settings are kept consistent with NC (Wang et al., 2019).

PT-RED (Xiang et al., 2022). We follow the default settings used in its original paper.

PT-RED-CSO. We set $\lambda = 0.1$ in Eq. 9. All remaining settings are kept consistent with PT-RED (Xiang et al., 2022).

MMBD (Wang et al., 2024b). We follow the default settings used in its original paper.

MMBD-CSO. We set $\lambda = 400$ in Eq. 6. All remaining settings are kept consistent with MMBD (Wang et al., 2024b).

MLBD-CSO. We set $\lambda = 400$ in Eq. 7. All remaining settings are kept consistent with MMBD (Wang et al., 2024b), except that we replace the maximum margin objective with the maximum logit objective.

A.3. Settings for Backdoor Attacks

In this section, we detail the backdoor attacks considered in our experiments, including the poisoning rate, clean test accuracy, and, for one-to-one and multi-to-one attacks, collateral damage.

BadNets (Gu et al., 2019). We consider a 3×3 random patch for CIFAR-10 and GTSRB and a 5×5 random patch for TinyImageNet, with a randomly selected location, which is fixed for all trigger images for a given attack.

Chessboard (Xiang et al., 2022). We consider a global additive perturbation (with size $3/255$) resembling a chessboard.

1-pixel (Tran et al., 2018). For CIFAR-10 and GTSRB, we consider a 1-pixel additive backdoor pattern which perturbs a single, randomly selected pixel by $75/255$ in all color channels, with this pixel location fixed for all trigger images for a given attack. For TinyImageNet, we consider a 4-pixel pattern that perturbs 4 randomly selected pixels.

Blend (Wang et al., 2019). We consider a 3×3 local random patch trigger with a blend ratio of $\alpha = 0.2$, with a randomly selected and fixed location for each image in a given attack.

WaNet (Nguyen & Tran, 2021). We consider a warping-based trigger with the default settings in the original paper.

IAD (Nguyen & Tran, 2020). We consider an input-aware dynamic trigger that perturbs each image using a small,

input-dependent noise pattern, following the settings in the original paper.

LC (Turner et al., 2019). We use projected gradient descent (PGD) to make adversarial samples and set the maximum perturbation size $\epsilon = 8$.

Bpp (Wang et al., 2022b). We consider a trigger that utilizes image quantization and dithering techniques with the default settings in the original paper.

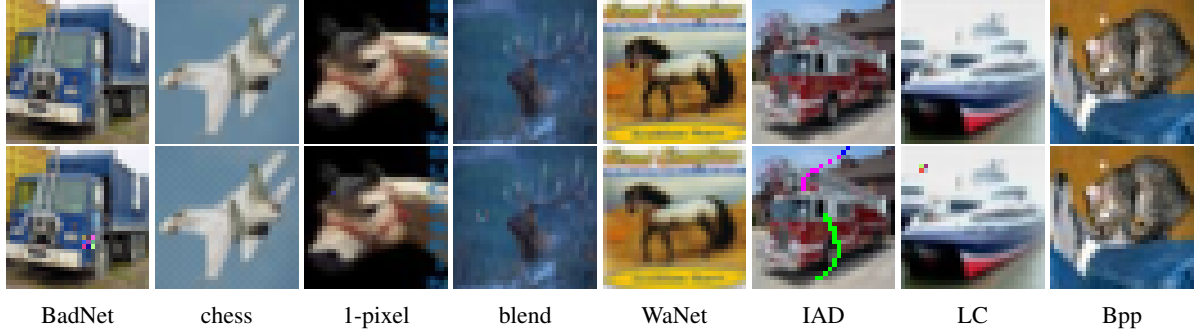


Figure 2. Example benign images (top) and corresponding poisoned images (bottom) used in our experiments.

In Figure 2, we demonstrate examples of poisoned images generated by different attacks on CIFAR-10.

Tables 8, 9, 10, 11, 12 and 13 give the poisoning rate (PR) for each attack (or for ML backdoors the dirty-label poisoning rate (DPR)), the average attack success rate (ASR) as well as the clean test accuracy (ACC) over the 10 clean models and 10 attacked models. *We set the poisoning rate to the minimum required to achieve a high attack success rate ($\approx 90\%$).* This choice makes the attack substantially harder to detect than the attacks (with much higher poisoning rates) used in the original defense papers, which explains the observed gap in detection accuracy.

For ML backdoors, the clean poisoning rate (CPR) equals the DPR. For one-to-one attacks on CIFAR-10 and seven-to-one attacks on GTSRB and TinyImageNet, we also show the average collateral damage (CD), which is calculated as the ratio of the non-source (and non-target) class samples, with trigger embedded, that are misclassified to the target class divided by the number of non-source (and non-target) class samples. Note that ML backdoor attacks have much lower CD than the non-ML backdoors.

Note also that DPR here is defined as the ratio of the number of dirty-label poisoned training samples from the source class to the total number of training samples. This definition ensures compatibility with PR, maintaining consistency in how poisoning levels are measured across different attack setups.

Table 8. On CIFAR-10, the poisoning rate (PR), average attack success rate (ASR) and average clean test accuracy (ACC) over the ensemble of 10 clean models and 10 attacked models for all-to-one backdoors.

CIFAR-10									
All-to-one	clean	BadNet	chess	1-pixel	blend	WaNet	IAD	LC	Bpp
PR(%)	0	0.1	0.5	0.5	5.0	5.0	5.0	5.0	1.0
ASR(%)	0	99.92	99.94	91.28	96.86	84.46	91.68	88.52	99.58
ACC(%)	91.61	91.59	91.06	91.12	91.48	90.72	91.23	91.77	91.00

A.4. Settings for model training

For BadNet, chessboard, 1-pixel and blend attacks, we adopt ResNet-18 (He et al., 2016) for CIFAR-10, MobileNet (Howard et al., 2017) for GTSRB, and ResNet-34 (He et al., 2016) for TinyImageNet. For LC, we adopt ResNet-18 for CIFAR-10 and PreActResNet-18 (Yu et al., 2018) for GTSRB and TinyImageNet. For WaNet, IAD and Bpp, we adopt PreActResNet-18 for all datasets. We train all DNNs with the Adam optimizer, using a learning rate of 0.001 and a batch size of 128, for 100 epochs on each dataset.

Improving the Sensitivity of Backdoor Detectors via Class Subspace Orthogonalization

Table 9. On CIFAR-10, PR (or for ML backdoors the dirty-label poisoning rate (DPR)), ASR, average collateral damage (CD) and ACC over the ensemble of 10 clean models and 10 attacked models for one-to-one backdoors.

CIFAR-10												
One-to-one	clean	BadNet	ML-BadNet	chess	ML-chess	1-pixel	ML-1pixel	blend	ML-blend	WaNet	IAD	Bpp
PR/DPR(%)	0	0.1	0.1	0.2	0.2	3.0	3.0	2.0	2.0	5.0	3.0	1.0
ASR(%)	0	92.16	92.11	90.53	93.98	93.01	92.01	90.52	91.67	90.96	87.04	100.00
CD(%)	0	54.27	24.55	58.36	24.44	36.38	4.26	41.75	6.48	3.64	25.99	2.52
ACC(%)	91.61	91.84	90.85	91.43	90.22	92.68	91.63	91.32	90.47	90.60	90.41	91.61

Table 10. On GTSRB, PR, ASR and ACC over the ensemble of 10 clean models and 10 attacked models for all-to-one backdoors.

GTSRB										
All-to-one	clean	BadNet	chess	1-pixel	blend	WaNet	IAD	LC	Bpp	
PR(%)	0	0.1	1.1	1.1	0.5	5.0	5.0	3.8	1.0	
ASR(%)	0	97.90	97.32	90.76	98.09	98.18	98.26	86.63	81.21	
ACC(%)	95.27	95.06	95.04	94.91	95.16	95.09	95.36	84.26	95.94	

Table 11. On GTSRB, PR/DPR, ASR, CD and ACC over the ensemble of 10 clean models and 10 attacked models for seven-to-one backdoors.

GTSRB									
Seven-to-one	clean	BadNet	ML-BadNet	chess	ML-chess	1-pixel	ML-1pixel	blend	ML-blend
PR/DPR(%)	0	3.2	3.2	8.1	8.1	3.2	3.2	3.2	3.2
ASR(%)	0	100.00	99.14	96.80	96.88	96.99	96.43	93.77	95.77
CD(%)	0	98.84	6.54	16.33	1.04	85.75	4.09	10.88	1.59
ACC(%)	95.27	95.58	93.12	94.93	94.26	95.27	96.43	94.64	94.52

Table 12. On TinyImageNet, PR, ASR and ACC over the ensemble of 10 clean models and 10 attacked models for all-to-one backdoors.

TinyImageNet									
All-to-one	clean	BadNet	chess	1-pixel	blend	WaNet	IAD	Bpp	
PR(%)	0	1.0	3.0	5.0	10.0	5.0	5.0	1.0	
ASR(%)	0	95.69	90.83	89.08	95.17	98.57	91.33	89.1	
ACC(%)	66.46	66.59	65.81	65.53	65.05	63.88	63.79	62.77	

Table 13. On TinyImageNet, PR/DPR, ASR, CD and ACC over the ensemble of 10 clean models and 10 attacked models for seven-to-one backdoors.

TinyImageNet									
Seven-to-one	clean	BadNet	ML-BadNet	chess	ML-chess	1-pixel	ML-1pixel	blend	ML-blend
PR/DPR(%)	0	1.8	1.8	5.3	5.3	5.3	5.3	14.0	14.0
ASR(%)	0	98.61	85.21	97.77	84.74	93.60	84.69	90.52	86.05
CD(%)	0	90.43	34.14	91.22	25.00	82.75	28.95	83.28	20.12
ACC(%)	66.46	65.37	64.70	64.52	64.83	64.20	64.94	65.98	66.07

B. Empirical Analysis of Feature Overlap

In this section, we provide an empirical analysis of feature overlap in backdoored models. We examine two complementary aspects: (i) trigger–intrinsic overlap, which measures how similar backdoor-induced features are to the intrinsic features of the target class across different network layers, and (ii) cross-class intrinsic overlap, which quantifies the inherent feature similarity between source classes and the target class. Together, these analyses help assess the separability of backdoor features from intrinsic features of a target class and provide empirical support for the design assumptions underlying CSO.

B.1. Quantitative Results on Layerwise Trigger-Intrinsic Feature Overlap between Backdoor Features and Target Class Intrinsic Features

In this section, we quantify “trigger–intrinsic” overlap, i.e., the separability of trigger features from intrinsic target class features, and investigate the overlapping layer-wise. To be specific, the overlap can be experimentally assessed by measuring the average rectified, masked feature cosine similarity, in various layers, between samples from non-target classes that contain the backdoor trigger and samples from the target class. We refer to this as the trigger-intrinsic feature overlap. Table 14 shows that the correlations between samples (without the trigger) from the target class – target class intrinsic feature overlap – are larger than the correlations between source class samples with the trigger and target class samples (without the trigger). This is as one would expect, and is supportive of CSO’s key assumption – that the intrinsic feature overlap between source class samples with the trigger and target class samples is low. In terms of the layer, we can see that, for most of the attacks (excepting IAD and Bpp), the trend is that the correlations in deeper layers (13th and 17th) are smaller than in the earlier layers. This gives some empirical support to our choice of a deep layer for measuring the CSO penalty.

B.2. Quantitative Results on Cross-Class Feature Overlap between Source Class Features and Target Class Intrinsic Features

In this section, we discuss intrinsic feature overlap across classes. This overlap can be experimentally understood by measuring, for different attacks, average rectified, masked feature cosine similarity between samples from source classes and samples from the target class. We show these results for CIFAR-10 in Table 15. We also evaluate the target class intrinsic feature overlap, i.e., average rectified, masked feature cosine similarity between target class samples as a baseline for comparison. The results show that, while the source-target correlations are smaller than the target-target correlations, there are significant source-target correlations under different attacks. This is in fact not so surprising, since different CIFAR-10 classes share high-level attributes. For example, dogs, cats, horses, and frogs all have eyes, and they all have legs. Despite these “feature overlaps”, MMBD-CSO is achieving strong detection results, and much better than existing baselines, against an array of attacks on this (CIFAR-10) data set.

C. More Details on Adaptive Attacks

In this section, we provide more details on the adaptive attack considered in Section 3.4. We additionally adopt another adaptive attack, Adaptive-Blend-2 (Qi et al., 2023) which diminishes the latent separation between clean and poisoned samples, following (Xu et al., 2024a). Both adaptive attacks aim to undermine the assumption that class-specific feature decoupling can cleanly separate backdoor and intrinsic features.

Adaptive-Blend. As described in Section 3.4, since our CSO variants rely on isolating backdoor information from target-class intrinsic features, we design an attack that explicitly leverages intrinsic content. Figure 3 shows an illustration of Adaptive-Blend. As shown in Table 5, CSO remains highly effective under Adaptive-Blend even up to a blend ratio of 1.0, highlighting its robustness against adaptive strategies.

Adaptive-Blend-2 (Qi et al., 2023). This attack generates partial-blend patterns as triggers for model training, and additionally includes regularized training samples in which backdoor patterns are embedded but labeled correctly, thereby circumventing defenses that rely on latent separability. We followed the original settings in the paper and generated 10 models. As seen in Table 16, our defense performs very well against Adaptive-Blend-2.

D. Results of Additional Ablation Studies

D.1. Ablations on MMBD-CSO

Table 14. Layerwise trigger–intrinsic feature overlap and target-class intrinsic feature overlap for different backdoor attacks.

Attack	Layer	Trigger-Intrinsic Feature Overlap	Target Class Intrinsic Feature Overlap
BadNet	5 th	0.38	0.46
	9 th	0.26	0.32
	13 th	0.21	0.33
	17 th	0.32	0.56
chess	5 th	0.40	0.40
	9 th	0.30	0.32
	13 th	0.21	0.31
	17 th	0.29	0.50
1-pixel	5 th	0.40	0.42
	9 th	0.31	0.36
	13 th	0.28	0.39
	17 th	0.27	0.71
blend	5 th	0.39	0.45
	9 th	0.26	0.32
	13 th	0.22	0.34
	17 th	0.21	0.52
WaNet	5 th	0.40	0.44
	9 th	0.48	0.53
	13 th	0.53	0.56
	17 th	0.33	0.61
IAD	5 th	0.23	0.26
	9 th	0.49	0.52
	13 th	0.53	0.59
	17 th	0.58	0.74
LC	5 th	0.40	0.51
	9 th	0.26	0.33
	13 th	0.17	0.28
	17 th	0.22	0.77
Bpp	5 th	0.17	0.28
	9 th	0.53	0.55
	13 th	0.46	0.58
	17 th	0.63	0.73

Table 15. Source class feature and target class intrinsic feature overlap, and target class intrinsic feature overlap for different backdoor attacks.

Attack→	BadNet	chess	1-pixel	blend	WaNet	IAD	LC	Bpp
Source Class, Target Class Intrinsic Feature Overlap	0.39	0.31	0.29	0.28	0.34	0.24	0.29	0.26
Target Class Intrinsic Feature Overlap	0.56	0.50	0.71	0.52	0.61	0.74	0.77	0.73

Table 16. Results on Adaptive-Blend-2.

Attack	ACC(%)	ASR(%)	DA(%)
Adaptive-Blend	91.73	90.56	98

In this section, we present additional ablation studies for MMBD-CSO on CIFAR-10 under one-to-one BadNet and ML-BadNet attacks.

Effect of N_{img} . In Table 17, we show the detection results when the number of available clean images is varied. Good results are achieved even with $N_{\text{img}} = 5$.

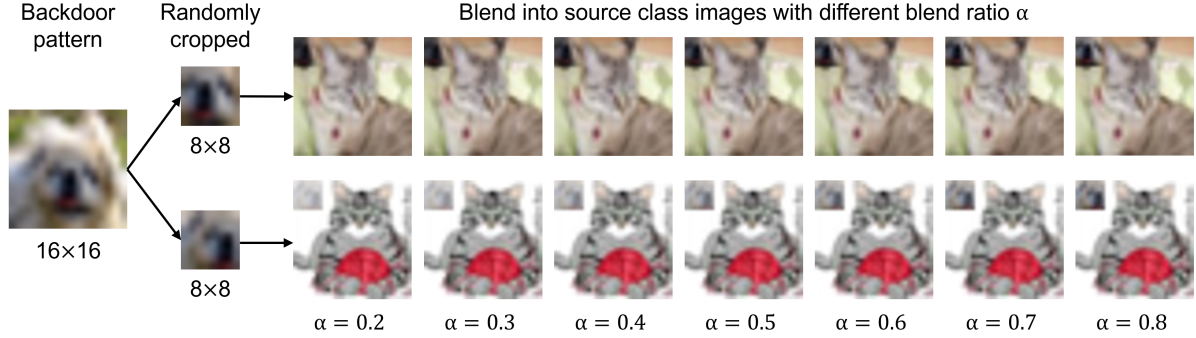


Figure 3. Illustration of our adaptive attack Adaptive-Blend. The target class is ‘dog’ and the backdoor trigger is a 16×16 dog’s face. To poison the training set, we generate sample-specific triggers by randomly cropping 8×8 regions from the original 16×16 dog-face pattern and blending them into source-class images. We considered attacks with blend ratios from 0.2 all the way up to 1.0. The figure is showing the blending into two different source class (cat) images. Note, particularly in the upper row, that the dog features become quite visible as the blend ratio is increased, i.e. the attack is becoming less “stealthy”.

Effect of Class-specific decoupling. In Table 18, we demonstrate that class-specific feature masking is essential for CSO to achieve effective detection. When a single global mask is estimated across all classes, detection performance drops for both BadNet and, especially, for ML-BadNet.

Effect of ReLU function in Eq. 5. In Table 19, we show the necessity of the ReLU function acting on the cosine similarity.

Table 17. Detection accuracies for different number of clean samples.

Attack $\downarrow$, $N_{img} \rightarrow$	1	5	10	50	100
clean	70	92	96	94	100
BadNet	84	96	96	98	96
ML-BadNet	44	68	70	66	72

Table 18. Detection accuracies with non-class-specific and class-specific feature masking.

Attack	non-class-specific	class-specific
clean	98	96
BadNet	88	96
ML-BadNet	36	70

Table 19. Detection accuracies without or with the ReLU function applied to the CSO penalty.

Attack	w/o ReLU	w/ ReLU
clean	96	96
BadNet	70	96
ML-BadNet	32	70

We further study the robustness of MMBD-CSO to potential contamination of the small clean dataset available to the defender. Specifically, we consider two realistic sources of degradation: label noise and domain shift in the clean set, and evaluate the resulting impact on MMBD-CSO’s detection performance.

Effect of Clean Data Quality regarding Label Noise. In Table 20, we consider mislabeling up to 30% of the clean set samples used by MMBD-CSO. The results show that MMBD-CSO is robust against such “label noise”.

Table 20. Detection accuracies on clean data with different mislabeling fractions.

Mislabeling Fraction	0%	10%	20%	30%
BadNet	96	96	90	90
ML-BadNet	70	72	68	66

Effect of Clean Data Quality regarding Domain Shift. In Table 21, we conduct two types of domain shift wherein the clean samples are altered to introduce either color jitter or Gaussian blurring. The results show that our method is fully robust to both types of domain shift.

Table 21. Detection accuracies on clean data with domain shift.

Domain shift	w/o	Color jitter	Gaussian blur
BadNet	96	94	96
ML-BadNet	70	70	74

D.2. Additional Ablations on Clean Data with Different Size and Quality

CSO does not introduce additional clean data requirements if the original detector needs a small clean dataset already. In particular, NC and PT-RED both rely on a clean support set for detection. Therefore, the effect of varying the number of clean images, as well as the effect of data quality such as those brought by noise and domain shift, should be evaluated not only for CSO but also for the original detector.

To further evaluate CSO’s sensitivity to quality and size of the clean support set, we now conduct experiments by varying the number of clean samples per class (N_{img}) and report results for NC, NC-CSO, PT-RED, and PT-RED-CSO under clean, BadNet, and ML-BadNet models. As shown in Table 22, CSO consistently improves detection performance across different clean set sizes. For example, NC-CSO yields consistent gains on ML-BadNet across all N_{img} , and CSO reduces false positives on clean models for both NC and PT-RED. In Table 23,24, we further evaluate robustness to noise and domain shift on the clean support set under BadNet. These results show that CSO improves stability under degraded clean data quality. For example, under Gaussian blur, NC drops to 2% detection accuracy, while NC-CSO maintains 44%.

These additional ablations across different CSO variants demonstrate that CSO remains stable under variations in both the size and quality of the clean support set.

Overall, CSO introduces only modest clean-data requirements for data-free detectors such as MMBD, and does not impose additional clean-data requirements for detectors that already rely on clean support sets. The latter setting is common in post-training detection methods. Furthermore, CSO consistently improves detection performance across different clean-data conditions while remaining robust to noise and domain shift.

These results suggest that CSO serves as a practical plug-and-play enhancement to existing detectors, offering consistent performance gains with little or no additional clean-data cost, depending on the base detector.

E. More Details on Time Complexity, Memory Usage and Computing Platform

In Table 25, we provide wall-clock times and memory usage for each detector on CIFAR-10 and TinyImageNet datasets. Note that, after deploying CSO, MMBD-CSO is still 22.24 $\times$, 2.87 $\times$, 2.46 $\times$, 7.79 $\times$ faster than UNICORN, BTI-DBF, NC and PT-RED respectively on CIFAR-10.

We further demonstrate the tradeoff between the computational overhead and performance gains brought by CSO. The computational overhead is measured as the fraction of increased computational time brought by CSO over the original MMBD computational time, averaged across attacks and models. The performance gain is measured as the fraction of detection accuracy improvement brought by CSO over the original MMBD detection accuracy, averaged across attacks and models. Please note that, when MMBD shows 0% detection accuracy in some attacks (e.g. Table 1, ML-chess, WaNet, Bpp), relative performance gain is mathematically undefined as the denominator is 0. In this case, we use the absolute gain, which is the detection accuracy improvement brought by CSO.

Table 22. Detection accuracies for different number of clean samples for NC, NC-CSO, PT-RED and PT-RED-CSO.

Attack	Detector	$N_{img} \rightarrow$	1	5	10	50	100
clean	NC		54	62	72	66	66
	NC-CSO		60	64	70	70	68
	PT-RED		8	36	76	70	62
	PT-RED-CSO		20	38	80	72	72
BadNet	NC		6	50	94	98	98
	NC-CSO		10	82	90	96	100
	PT-RED		2	2	6	6	10
	PT-RED-CSO		8	6	8	18	12
ML-BadNet	NC		0	30	48	52	66
	NC-CSO		4	38	66	76	84
	PT-RED		10	8	10	10	12
	PT-RED-CSO		12	12	10	16	16

Table 23. Detection accuracies on clean data with different mislabeling fractions for NC, NC-CSO, PT-RED and PT-RED-CSO.

Detector ↓	0%	10%	20%	30%
NC	94	90	72	70
NC-CSO	90	92	84	80
PT-RED	10	10	6	0
PT-RED-CSO	12	14	10	2

The computational overhead of MMBD-CSO over MMBD on CIFAR-10 across attacks is 65%, and the performance gain on CIFAR-10 across attacks is 169%. This leads to an efficiency ratio (performance gain over computational overhead) of 2.6. The results again demonstrate that CSO is a practical, economic and strongly effective add-on to existing detectors which strikes a nice balance between computational overhead and performance improvement.

In this paper, all experiments are conducted on a single NVIDIA RTX 3090 Ti GPU using PyTorch.

F. Additional Experimental Results against More Attacks

In this section, we present additional experimental results for MMBD and MMBD-CSO on CIFAR-10 against other stealthy attacks that focus on latent feature preservation, clean label attacks, multi-trigger attacks and multi-target attacks. All attacks considered are all-to-one except the multi-target attack, which is N-to-N. We ran MMBD and MMBD-CSO on each model 5 times.

F.1. Results on More Stealthy Attacks that Preserve Latent Features

In this section, we present more rigorous empirical studies against backdoor threats that could potentially undermine the CSO defense, namely latent feature separation-focused attacks. Specifically, these attacks are designed to reduce the embedding-space discrepancy between backdoored samples and clean samples. As a result, they pose a particular challenge to CSO, since they explicitly attempt to camouflage the backdoor’s footprint in the latent feature space on which CSO operates.

Despite this, our results show that MMBD-CSO consistently and stably improves upon the baseline detector, even under these highly stealthy and adversarially designed attacks.

F.1.1. RESULTS ON DRUPE ATTACK (TAO ET AL., 2024)

The DRUPE (Tao et al., 2024) attack is designed to make out-of-distribution, highly clustered poisoned samples blend into the clean data distribution and also to be more dispersed.

Table 24. Detection accuracies on clean data with domain shifts for NC, NC-CSO, PT-RED and PT-RED-CSO.

Detector ↓	w/o	Color Jitter	Gaussian Blur
NC	94	84	2
NC-CSO	90	86	44
PT-RED	10	10	4
PT-RED-CSO	12	12	10

Table 25. Runtime and VRAM consumption comparison across different backdoor detection methods.

Metric	Dataset	UNICORN	BTI-DBF	NC	NC-CSO	PT-RED	PT-RED-CSO	MMBD	MMBD-CSO
Time (s)	CIFAR-10	1401	181	155	199	491	687	38	63
	TinyImageNet	–	407	562	786	1230	1498	538	782
VRAM (MiB)	CIFAR-10	–	–	592	776	634	926	542	774
	TinyImageNet	–	–	1496	2308	8662	13730	1692	2542

Settings. We reproduced the DRUPE attack with its official code under the default settings, where the encoder is pretrained on CIFAR-10 and downstream classifiers are trained on GTSRB. We generated five backdoored downstream models with 2 different reference inputs for each model.

Results. MMBD achieved a detection accuracy of only 4%, while MMBD-CSO achieved 60%, demonstrating a huge improvement over the baseline MMBD method for this challenging, advanced attack.

F.1.2. RESULTS ON BYPASSING ATTACK (SHOKRI ET AL., 2020)

The Bypassing Attack (Shokri et al., 2020) is designed to maximize the indistinguishability of the latent representations of poisoned data and clean data.

Settings. We implemented the Bypassing attack (Shokri et al., 2020) on ten BadNet baseline models with randomly chosen target labels and then jointly trained the backdoored model with the discriminator network to suppress feature differences between backdoored and clean samples, following the experimental configuration described in the original paper.

Results. MMBD achieved a detection accuracy of 18%, while MMBD-CSO achieved 56%, demonstrating a huge improvement over the baseline MMBD method on this challenging attack.

F.2. Results on More Clean-Label Attacks

In this section, we evaluate our methods on more types of clean label attacks. To be specific, we consider Refool (Liu et al., 2020), Narcissus (Zeng et al., 2023), and SIG (Barni et al., 2019) attacks. Refool creates triggers by simulating realistic environment reflections on objects. Narcissus implants a backdoor signature directly into the model’s representation space by learning a perturbation for some training samples so that their hidden-layer activations become closer to a specific target signature. SIG generates triggers using a global sinusoidal perturbation.

Settings. We follow the default settings used in the original papers. 10 models for each attack are generated with randomly chosen target class.

Results. In Table 26, we show that MMBD-CSO gives DA improvement over MMBD across these attacks.

Table 26. Detection accuracies for more clean label backdoor attacks.

CIFAR-10	Refool	Narcissus	SIG
MMBD	76	16	100
MMBD-CSO	94	56	100

Table 27. Detection accuracies on CIFAR-10 for BAN and MMBD-CSO across different attack types.

Detector	N_img	clean	BadNet	ML-BadNet	chess	ML-chess	1-pixel	ML-1-pixel	blend	ML-blend	WaNet	IAD	Bpp
BAN	500	100	82	50	38	36	66	46	62	32	10	20	34
MMBD-CSO	10	96	96	70	78	66	100	94	84	78	20	48	36

F.3. Results on Multi-Trigger Attack (Li et al., 2024)

(Li et al., 2024) propose a multi-trigger attack MTBA where multiple adversaries use different types of triggers to poison the same dataset.

Settings. We considered all-to-one MTBA with 10 different triggers and followed the default settings used in the original paper. 10 backdoored models are trained with a randomly chosen target class.

Results. MMBD achieved 90% detection accuracy, while MMBD-CSO achieved 98%, which demonstrates the CSO method’s resistance to multi-trigger attacks.

F.4. Results on Multi-Target Attack (Xue et al., 2023)

(Xue et al., 2023) propose an N-to-N backdoor attack that utilizes Discrete Cosine Transform (DCT) steganography to embed unique binary triggers into different RGB channels of an image, where each channel independently activates a different target label.

Settings. We considered N-to-N multi-target attack where different triggers and target labels are set for different RGB channels. We followed the default settings used in the original paper. 10 backdoored models are trained with randomly chosen target classes.

Results. MMBD achieved 86% detection accuracy, while MMBD-CSO achieved 100%, which demonstrates the CSO method’s resistance to multi-target attacks.

F.5. Results on TITIM Attack (Lin et al., 2025)

(Lin et al., 2025) vary the trigger intensity, such as opacity or size, and train backdoored models using triggers at different intensities and then evaluate them under inference triggers with different intensities. They prove that trigger intensity mismatch can improve both attack robustness and stealthiness

Settings. we vary the trigger intensity from 0.5 to 1.0 and create 10 TITIM-BadNet models for each trigger intensity. Note that we choose trigger intensity consistent with the original paper (see Table 7 in (Lin et al., 2025)) and in a range to make the attack successful ($ASR \geq 90\%$).

Results. Both MMBD and MMBD-CSO show 100% detection accuracy on these models.

G. Additional Experimental Results with Additional Baseline

In this section, we consider another recent post-training detector BAN (Xu et al., 2024b). Table 27 shows BAN has inferior performance to MMBD-CSO. We conjecture that competitor detectors were not designed to address very low poisoning rate attacks, leading to performance highly impacted by the “lowest feasible poisoning rates” used in our experiments.

H. Additional Experimental Results with More Model Architectures

In this section, we present additional experiments using more representative model architectures. Specifically, we evaluate our methods on CIFAR-10 using VGG-16 (Simonyan & Zisserman, 2015) and ViT (Dosovitskiy et al., 2021) (without pre-training) under both BadNet and ML-BadNet attacks. As shown in Table 28, MMBD-CSO achieves substantially higher detection accuracy compared with MMBD. These results further demonstrate the generalizability and robustness of our proposed approach across diverse neural network architectures.

I. Results on WaNet, IAD, LC and Bpp attacks with Higher Poisoning Rate

Table 28. Detection accuracies on larger model architectures.

Model	Attack	MMBD	MMBD-CSO
VGG-16	BadNet	68	92
	ML-BadNet	0	36
ViT	BadNet	42	78
	ML-BadNet	0	20

In this section, we provide additional context explaining the relatively low absolute detection accuracies reported in the main paper under certain stealthy attacks, and we examine how increasing the poisoning rate affects detection performance. The detection accuracies in the main paper for all methods are all “unusually” low because we chose a low poisoning rate (to make the detection problem as challenging as possible). When higher poisoning rates are chosen, all methods generally achieve higher true detection rates. Much higher poisoning rates were used in the original papers for the baseline methods, and much higher detection rates were reported in these original papers. We now demonstrate this for MMBD and MMBD-CSO in Table 29. Here, the poisoning rate was increased to 10% and the detection rates are much higher on WaNet, IAD, LC, and Bpp than the results reported in Table 1. At the same time, MMBD-CSO is still achieving a detection advantage over MMBD – at this higher poisoning rate, it is achieving excellent performance in detecting these stealthy attacks.

Table 29. Detection accuracies on CIFAR-10 under all-to-one WaNet, IAD, LC and Bpp attacks with higher poisoning rates of 10%.

CIFAR-10	WaNet	IAD	LC	Bpp
MMBD	98	88	78	66
MMBD-CSO	96	96	100	88

J. CSO with Backdoor Mitigation

Backdoor mitigation is another branch of backdoor defense that aims to neutralize compromised models such that they behave correctly even when presented with poisoned inputs. Existing backdoor mitigation methods include (Wang et al., 2019; 2024a; Li et al., 2021; Zhao et al., 2024; Arora et al., 2024; Li et al., 2026). In this section, to explore possible mitigation adaptations for CSO, we incorporate CSO into two representative post-training mitigation-based defenses for the image domain: NC and MMAC (Wang et al., 2024a). NC mitigates the backdoor by unlearning using the estimated mask, trigger pattern, and detected target class. MMAC is the mitigation extension of MMBD that clips ReLU activations to minimize the maximum achievable margin of each class. NC-CSO gives mitigation results by unlearning using the estimated mask, pattern, and detected target class given by NC-CSO detection. MMAC-CSO mitigates by adding the CSO penalty to its main objective (see Formula (4) in (Wang et al., 2024a)).

Table 30 reports mitigation results for the two methods. CSO improves both baselines in not only lower ASR, but also higher ACC (clean accuracy). Thus, CSO can help mitigation, as well as detection.

K. Varying the CPR for Mixed Label Attacks

Our mixed-label attack is designed to reduce unintended misclassification of trigger-bearing, non-source class samples to the target class. Below we assess how varying the clean-label poisoning rate (CPR) under a fixed dirty-label poisoning rate (DPR) impacts collateral damage and detection accuracy. For clarity and brevity we focus on ML-BadNet here; experiments with other mixed-label attacks show comparable trends.

As shown in Table 31, mixed-label BadNet significantly reduces collateral damage compared to the purely dirty-label BadNet, with collateral damage decreasing further as CPR increases. Notably, as CPR is varied, ML-BadNet preserves high backdoor attack success rates while leaving clean accuracy largely unaffected. In contrast, MMBD-CSO exhibits a clear drop in detection accuracy between BadNet and ML-BadNet, and the gap widens as CPR grows. These results reinforce that mixed-label attacks are substantially more difficult to detect than pure dirty-label attacks.

Table 30. Defense performance against BadNet and ML-BadNet attacks.

Detector	BadNet		ML-BadNet	
	ACC	ASR	ACC	ASR
No Defense	91.84	92.16	90.85	92.11
NC	88.79	15.98	87.37	19.09
NC-CSO	88.83	6.24	87.50	7.21
MMAC	88.02	2.54	87.98	3.92
MMAC-CSO	88.88	2.01	88.74	2.94

Table 31. Average ASR, CD, ACC and DA for different CPR under DPR = 0.1%.

DPR = 0.1%					
CPR(%)	0	0.05	0.10	0.20	0.30
ASR(%)	92.16	93.61	92.11	87.14	88.85
CD(%)	54.27	29.343	24.55	12.46	11.77
ACC(%)	91.84	91.07	90.85	90.62	90.89
DA(%)	96	76	70	62	46

L. More Discussion on Backdoor Effects On Class Logits

In this section, we provide additional analysis to better understand why MLBD-based detectors tend to underperform MMBD-based detectors, and why this performance gap persists even when both are augmented with CSO. In particular, we empirically examine how backdoor triggers affect class logits, with a focus on whether backdoor success primarily arises from boosting the target-class logit or from suppressing logits of other classes.

Our conjecture is that, in some backdoor settings, especially mixed-label attacks, the backdoor may not strongly increase the logit of the target class. Instead, it may operate largely by suppressing the logits of non-target classes, thereby increasing the classification margin without a correspondingly large increase in the target-class logit. Since MMBD explicitly exploits margin structure while MLBD focuses more directly on target-class logit behavior, this mechanism could explain the observed detection gap between MMBD-CSO and MLBD-CSO.

To investigate this effect, we analyze class logits under a one-to-one ML-BadNet attack on CIFAR-10, a setting where MMBD-CSO substantially outperforms MLBD-CSO. We compute the average logits across all classes for (i) clean samples from the source class and (ii) backdoored samples from the same source class with the trigger applied. In this example, the source class is class 0 and the target class is class 1. Figure 4 reports the average logits per class for both types ((i) and (ii)) of samples. In the presence of the backdoor trigger, the average target-class logit increases by around 16, and the average source-class logit decreases by a similar amount, effectively transferring decision confidence from the source class to the target class. Simultaneously, all non-target, non-source classes experience substantial logit suppression, with decreases ranging from approximately 6 to 16. These effects are not negligible.

Taken together, these results indicate that the backdoor’s effectiveness arises not solely from boosting the target-class logit, but also from simultaneously suppressing the logits of all competing classes. Consequently, the classification margin—which depends on the difference between the target logit and the largest non-target logit—can increase significantly even when the target class logit increase is moderate. This behavior provides empirical support for the conjecture that margin-based detectors (MMBD) are inherently better suited to capturing such backdoor effects than logit-based detectors (MLBD). Since margin gains exploit both target-class logit increases and widespread non-target logit suppression, MMBD-CSO can leverage a richer signal than MLBD-CSO in these settings.

M. Discussion on Limitations of the Method

As demonstrated in our results, CSO-based detectors consistently yield stable and substantial improvements over their respective baseline detectors. However, CSO does not yet achieve very strong absolute performance against certain highly

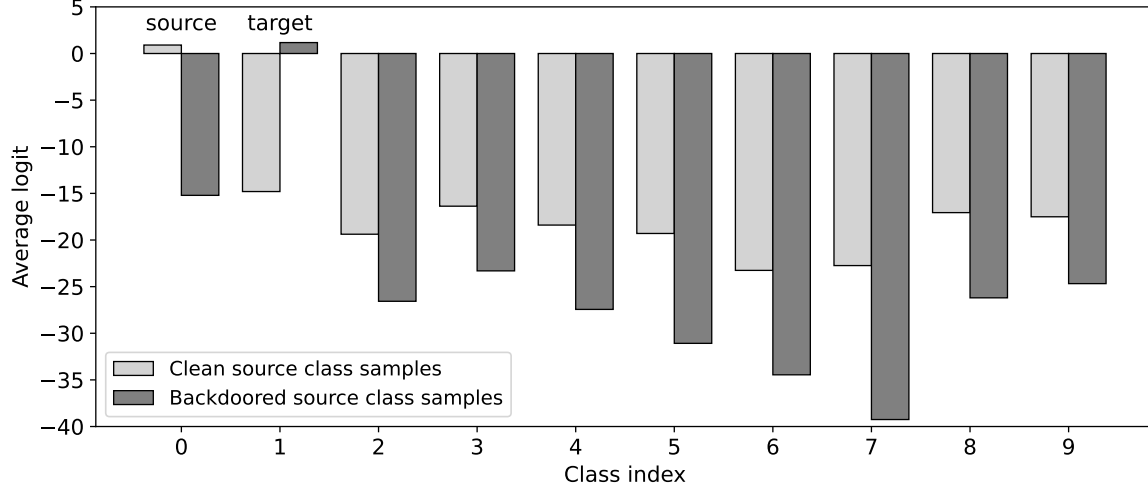


Figure 4. Average logits per class for clean and backdoored samples from the source class.

stealthy attack types (e.g. Bypassing attack). This indicates that there remains room for improvement, and motivates future research into strengthening CSO to more effectively counter such attacks.

Additionally, the current CSO framework is only deployed in the image domain and computer vision tasks. An important direction for future work is therefore to extend the CSO principle beyond vision applications and develop CSO-based detectors for domains such as large language models and multimodal systems.